Labor markets, distribution and the political economy of international trade

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Outline

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Motivation

- Normally think of tariffs benefiting firms, hurting consumers
  - Protected firms increase output, make higher profits
  - More jobs/higher wages for workers at protected firms
  - Consumers suffer from higher prices, lower consumption
- But firms are consumers too!
  - "National security" tariffs on steel & aluminum
  - "China tariffs" focused on intermediate & capital goods
- Offsetting effects on employment
  - Negative effects in downstream industries
  - Positive effects in protected industries

Motivation

- Last time large scale tariffs on key inputs?
  - Bush steel tariffs of 2002
    - "Safeguard" MFN tariffs
    - 13-30% tariffs on $\approx$ 170 8-digit steel products
  - Steel most important manuf input in US economy ▸ IO structure
- But, little evidence on employment effects of Bush steel tariffs
  - Most cited analysis: Baughman & Francois (2003)
    - Bush steel tariffs cost 200,000 jobs
    - International trade consulting firm for pro-trade lobby

Research question

- What effects did Bush steel tariffs have on local employment
  - ... in steel-consuming industries?
    - Manufacturing sector, subset of steel-intensive industries
  - ... in steel-producing industries?
- Locations = US commuting zones (CZs)
  - 722 piece "jigsaw puzzle" of US

Econometrics: thought experiment

- Split CZs into two groups
  - Treatment group: heavily vulnerable to Bush steel tariffs
  - Control group: not heavily vulnerable to Bush steel tariffs & IO structure Vulnerability
- Compare employment change pre vs post Bush tariffs ...
  - ... between treatment & control groups of CZs & Emp

Results

1. Large & persistent negative effects on steel-consuming emp
   - 2000-2003 mean CZ Δemp share:
     - -1.3% pts for manuf emp, -0.3% pts for steel cons emp
   - Compare CZs at 25th and 75th percentile of vulnerability
     - Δmanuf emp share of -0.5% pts ($\approx$ 40% of mean Δ)
     - ΔsteelCons emp share of -0.35% pts ($>$ 100% of mean Δ)
     - Effects peak at end of Bush tariffs, stable through 2008
2. Essentially, zero effects on steel-producing emp

- Policy implication: Trump (now Biden) tariffs...
  - Large long lasting damage on US manuf sector

Input-output structure

Most heavily used inputs (Avg r_ij)			Most steel-intensive industries (r_steel, j)		
NAICS (i)	Industry	Avg. Usage per $100	Industry (j)	Usage per $100 (TR)	Usage per $100 (DR)
420000	Wholesale trade	$8.46	Rolled steel shape manuf	$40.51	$39.36
550000	Mgmt of co's & enterp's	$5.71	Iron, steel pipe & tube	$38.24	$38.47
531000	Real estate	$3.47	Steel wire drawing	$36.34	$37.36
484000	Truck transportation	$3.09	Custom roll forming	$32.28	$31.23
331111	Iron & steel mills	$2.67	Prefab. metal buildings	$29.05	$26.92
211000	Oil & gas extraction	$2.64	Spring & wire manuf	$23.15	$24.54
221100	Power generation & supply	$2.27	Fab. pipe & pipe fitting manuf	$22.79	$21.71
324110	Petroleum refineries	$2.09	Iron & steel forging	$20.15	$18.34
325190	Other basic org chem manuf	$1.88	Metal tank manuf	$18.47	$16.63
3221A0	Paper & paperboard mills	$1.83	Elevator manuf	$18.40	$16.26

- Steel is crucial input in the economy

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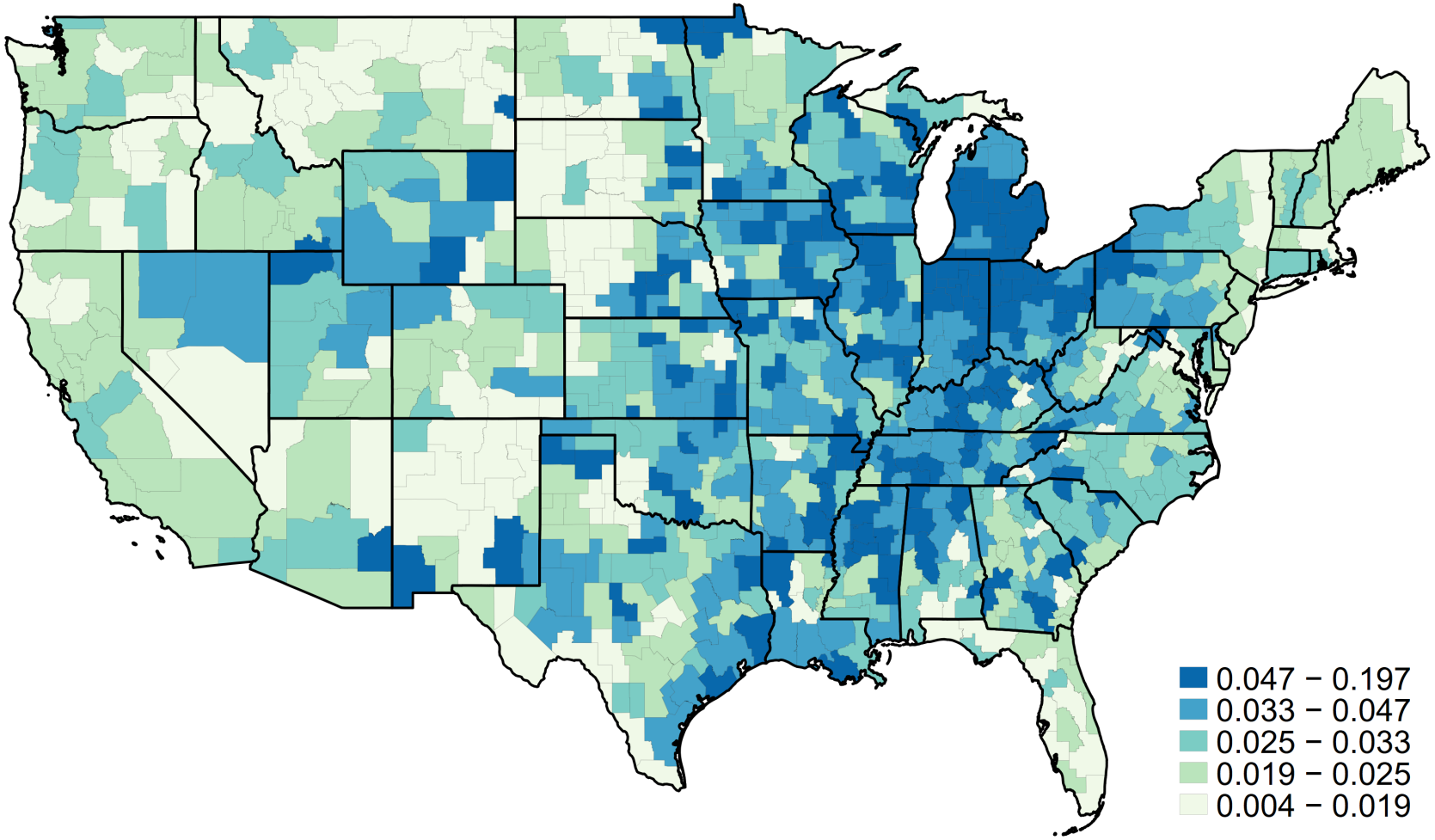
CZ vulnerability



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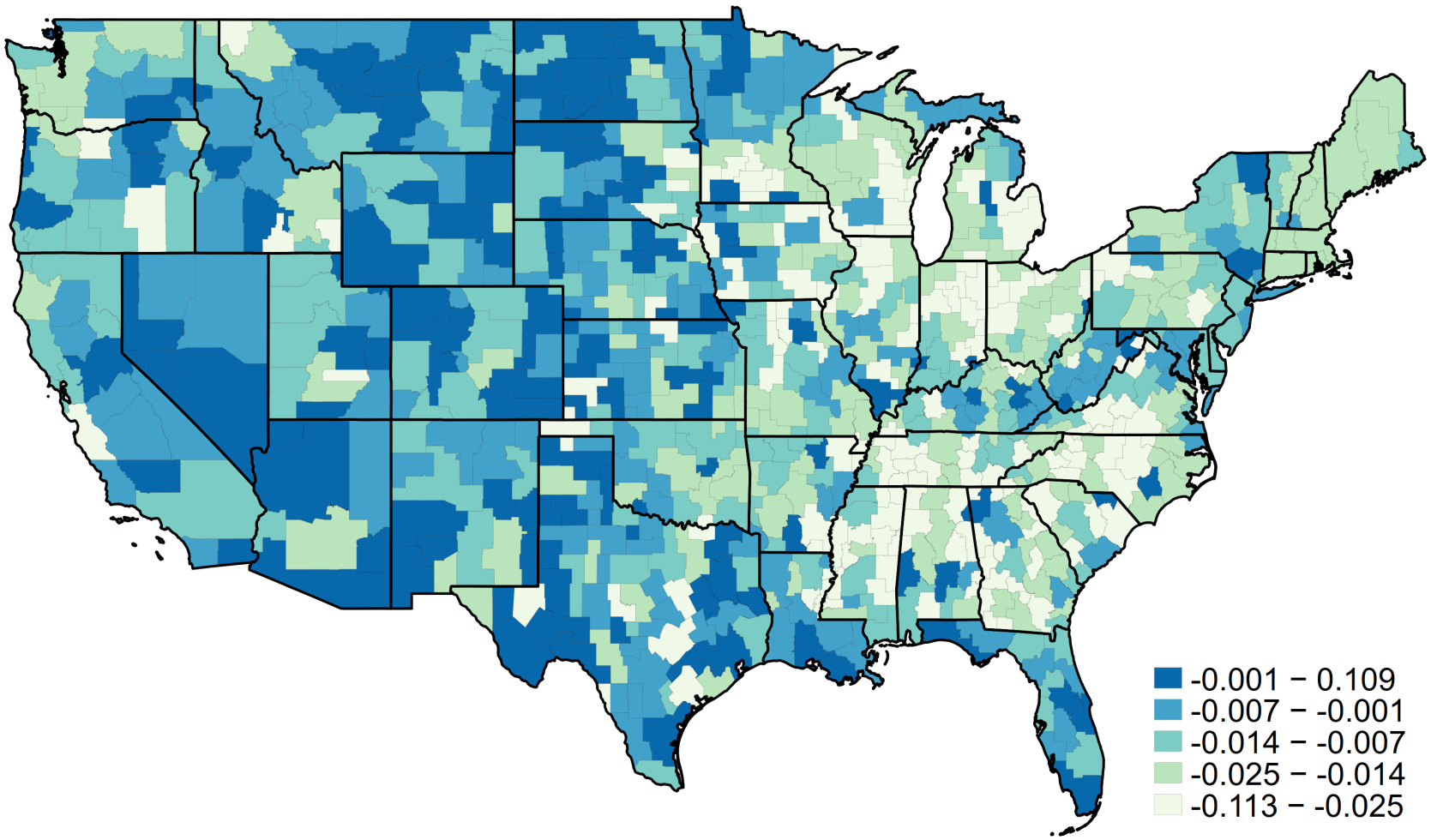
Changing CZ manufacturing employment share 2000-2003



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Vulnerability

Outline

Motivation

- Trump administration has imposed waves of tariffs
  - Section 201 "safeguard tariffs" on solar panels & washing machines
  - Section 232 "national security tariffs" on steel & aluminium
  - Section 301 "unfair trade practices tariffs" on China
- Theory says, in general, US consumers and foreigners pays the tariffs
  - Acoording to theory
    - US consumers pay some via higher prices in US
    - Foreign producers pay some via lower export prices
    - US welfare increases only if TOT gain $>$ DWLs
  - Trump claims Foreigners are paying
    - US can only gain if Trump is "sufficiently" correct

Trump's tariff waves

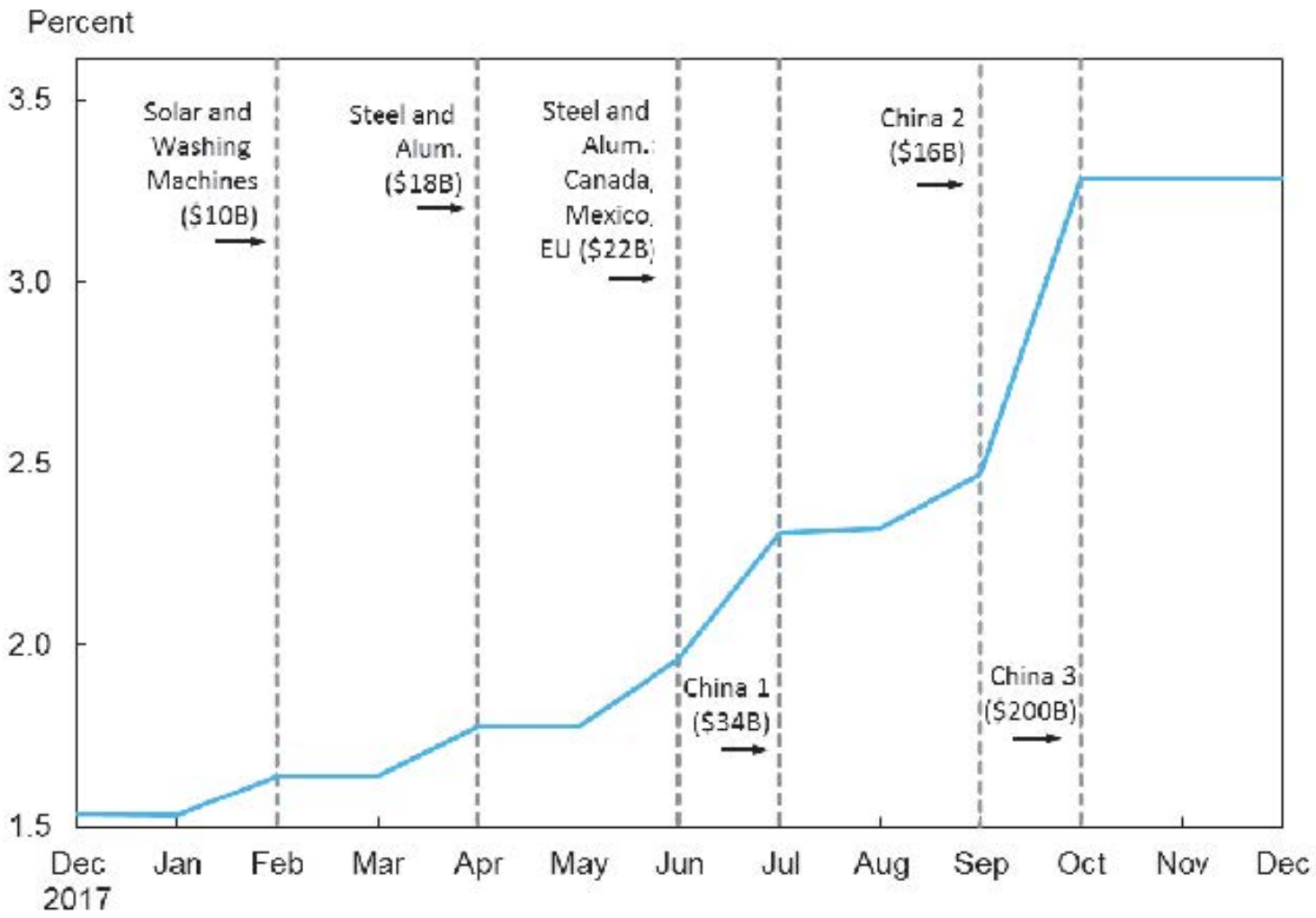


Figure: US trade war tariffs: 2018

Research question

- Empirically, who pays the tariffs?
  - US consumers or Foreign producers?

Econometrics: thought experiment

- Compare price change (pre vs post trade war) between treatment & control groups
- Control group
  - Imported products from countries not facing trade war tariffs
    - EX: Electrical equip from AUS, EU, CAN, MEX, KOR...
- Treatment group
  - Imported products from countries facing trade war tariffs
    - EX: Electrical equip from CHN

Graph: tariffs and domestic US prices

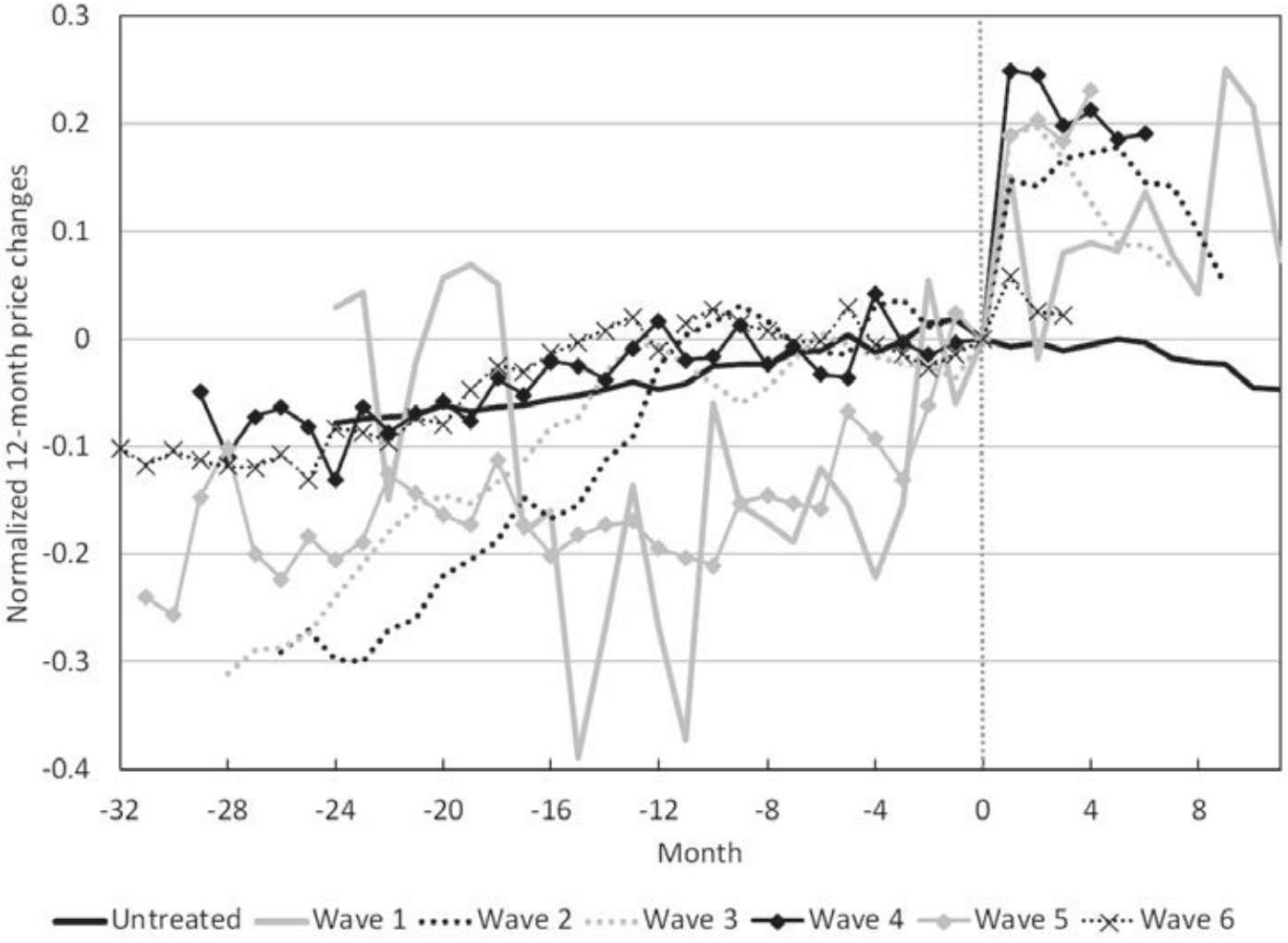


Figure: Trade war tariffs and domestic US prices: 2015-2018

Regression results: tariffs and domestic US prices

- Essentially, tariffs fully passed through to domestic US prices
  - Regression results: 1% $\uparrow$ tariff $\Rightarrow$ 0.01% $\downarrow$ foreign exporter price
  - No statistically significant effect on foreign export price

Graph: tariffs and US imports

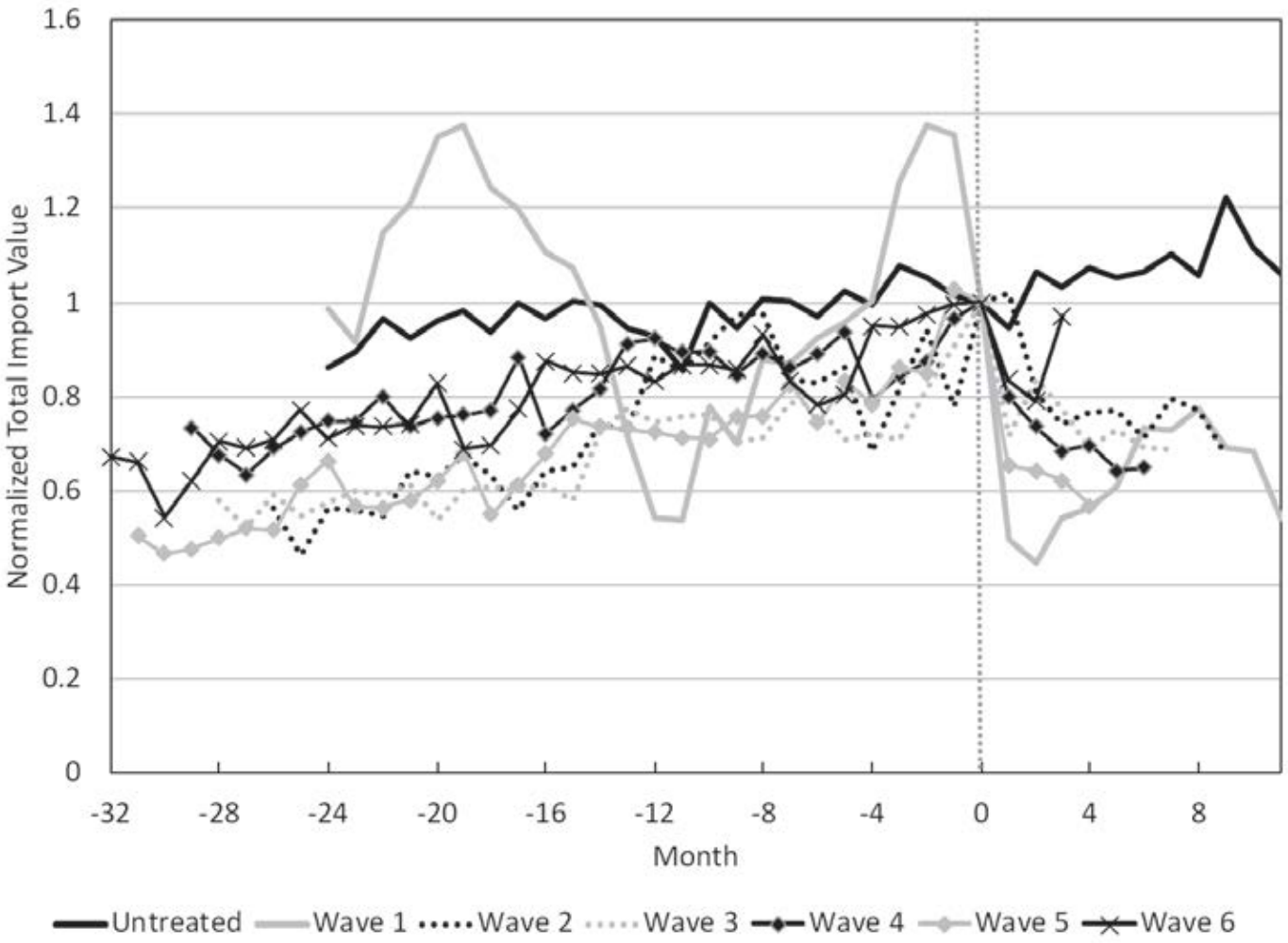


Figure: Trade war tariffs and US imports: 2015-2018

Regression results: tariffs and US imports

- Large impact on exports. Regression results say
  - 1% $\uparrow$ tariff $\Rightarrow$ 6.3% $\downarrow$ imports...
    - ... *relative* to unaffected products
  - This is \$132bn annually
    - $\approx$ 5% of annual US IM, $\approx$ 26% annual US EX to CHN

Regression results: deadweight losses & tariff revenue

- Effects of US trade war tariffs on US economy
  - 2018: DWL = \$8.2bn, TR = \$15.6bn
  - Per month, post Sept wave: DWL = \$1.4bn, TR = \$3.2bn
  - All of TR gain is transfer from consumers to govt. No TOT gain.

Regression results: foreign retaliatory tariffs

- No effect on US exporter prices (same as effects of US tariffs)
- Large impact on exports. Regression results say
  - $1\% \uparrow$ tariff $\Rightarrow$ $3.9\% \downarrow$ exports...
    - ... *relative* to unaffected products
  - By end of 2018: \$2.4bn per month, \$28.8bn annually
    - $\approx 2\%$ of annual US EX, $\approx 20\%$ annual US EX to CHN

Conclusion

- Consumers are paying the trade war tariffs
  - Foreign exporter prices unchanged
  - Applies to US tariffs and foreign retaliatory tariffs
- Large fall in imports due to tariffs
  - Applies to US imports and foreign imports (i.e. US exports)
- No effect on US exporter prices (same as effects of US tariffs)
  - Large impact on exports. Regression results say
    - $1\% \uparrow$ tariff $\Rightarrow 3.9\% \downarrow$ exports
    - US can be facing 5%, 10%, 20% foreign retaliatory tariffs...

Outline

Motivation

- WTO must treat other *WTO members* equally
  - Most Favored Nation (MFN) principle
    - Exception: Preferential Trade Agreements (e.g. NAFTA)
- US policy on non-WTO members
  - Grant "normal trade relations (NTR)" $\Rightarrow$ impose MFN tariffs
  - Don't grant NTR $\Rightarrow$ impose *very high* non-NTR tariffs
    - Non-NTR tariffs set in 1930

Motivation and research question (cont.)

- Pre-2001, US annually decided whether to grant NTR
  - From 1980, US granted NTR status to China every year
  - But... politically controversial during 1990s
- China entered WTO in 2001
- WTO entry $\Rightarrow$ permanent NTR (PNTR)
  - Effectively, US liberalizes tariff policy on China
  - Potential threat of very high non-NTR tariffs
    - depressed US imports from China
  - Threat now gone
    - import surge from China

Research question

- Effect of PNTR on US manuf employment?

Econometrics: thought experiment

- Look at change in industry employment pre and post PNTR
- Control group
  - US *industries* where non-NTR tariff = NTR tariff
- Treatment group
  - US *industries* where non-NTR tariff > NTR tariff

Need industry specific measure of "NTR gap"

Econometrics: regression

- 300 industries, annual data 1990-2007
- Estimate causal impact of PNTR on US manufacturing employment

Econometrics: results

- PNTR (i.e. elimination of $NTRgap_j$ in all industries) $\Rightarrow\downarrow$ manuf emp by 15%
  - *relative* to % change in manuf emp upon China entering WTO in hypothetical situation where NTR gap had always been zero in all sectors

Outline

Motivation & research question

Motivation

- Rapid rise of China
  - 1990-2007: massive $\uparrow$ US imports from China

Research question

- How does this affect US labor market outcomes?
  - Manufacturing employment, unemployment, labor force participation, wages

Econometrics: thought experiment

- Look at change in labor market outcomes 1990-2007
- Control group
  - US *locations* whose workers don't compete with China
- Treatment group
  - US *locations* whose workers compete with China

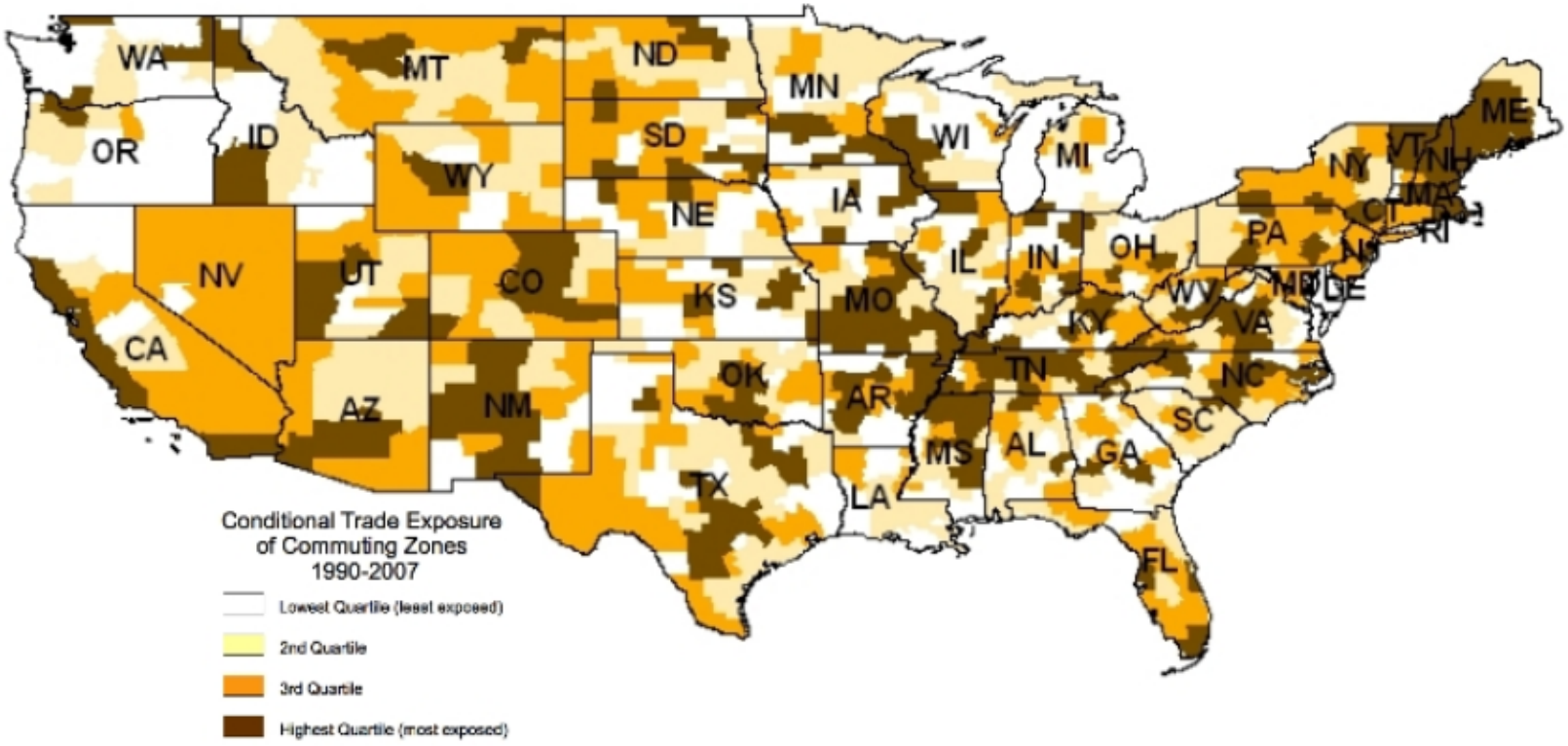
Need local measures of rising import penetration

Data: local import penetration

Local import penetration is rising a lot when...

- workers in *location c* concentrated in *industries* where Chinese import penetration growing rapidly

Example: local import penetration



Among 50 Largest Commuting Zones

(A) Largest Increase in Exposure

1. San Jose, CA
2. Raleigh, NC
3. Providence, RI

(B) Smallest Increase in Exposure

1. Detroit, MI
2. Grand Rapids, MI
3. Seattle, WA

Figure: Local Chinese import penetration

Main result I: effect on manuf jobs

- US manufacturing employment share
  - B/w 1990 and 2007 :$\downarrow$ 4.18% pts
  - China accounts for 22.5% of this 4.18%

Main result II: worker adjustment

How do workers adjust to less manuf jobs in highly exposed locations?

- Migrate to other parts of US?
  - No. Exposure doesn't affect migration patterns.
- Switch to non-tradable sector?
  - No. Non-tradable doesn't expand in highly exposed locations.
  - But $\downarrow$ wages here b/c $\uparrow$ competition for these jobs
- Rather, $\uparrow$ share of population...
  - unemployed and leaves labor force (incl. disability insurance)
  - Especially true if don't finish college

Outline

Motivation & research question

Motivation

- NAFTA very controversial in US
  - 1994: NAFTA comes into force
  - 2000: US has eliminated most tariffs on Mexico

Research question

- Effects of falling NAFTA tariffs on US wages?
  - When tariff for *own* industry falls
  - When tariffs for important *local* industries fall
  - Heterogeneity by worker education level?

Econometrics: thought experiments

- Two different thought experiments
  1. Compare wage growth 1990-2000 for two groups of workers
     - Treatment group: workers in high NAFTA exposure *locations*
     - Control group: workers in low NAFTA exposure *locations*
  2. Compare wage growth 1990-2000 for two groups of workers
     - Treatment group: workers in high NAFTA exposure *industries*
     - Control group: workers in low NAFTA exposure *industries*
- Actually do these two thought experiments separately for
  - high school dropouts
  - high school graduates
  - some college
  - college graduates

Need local measures of "local NAFTA exposure"

Local NAFTA exposure

When is local NAFTA exposure large?

- A location's workers are concentrated in *industries* where
  - Big fall in tariffs on Mexico and
  - Mexico has strong comparative advantage

Main result I: elimination of local tariffs

- Compare wage growth 1990-2000 for the following two workers
  - high school dropout in most exposed NAFTA location (NC, SC)
  - high school dropout in least exposed NAFTA location (DC, VA, MD)
- Former has 8% pt lower wage growth
  - Eg: 1% for former, 9% for latter

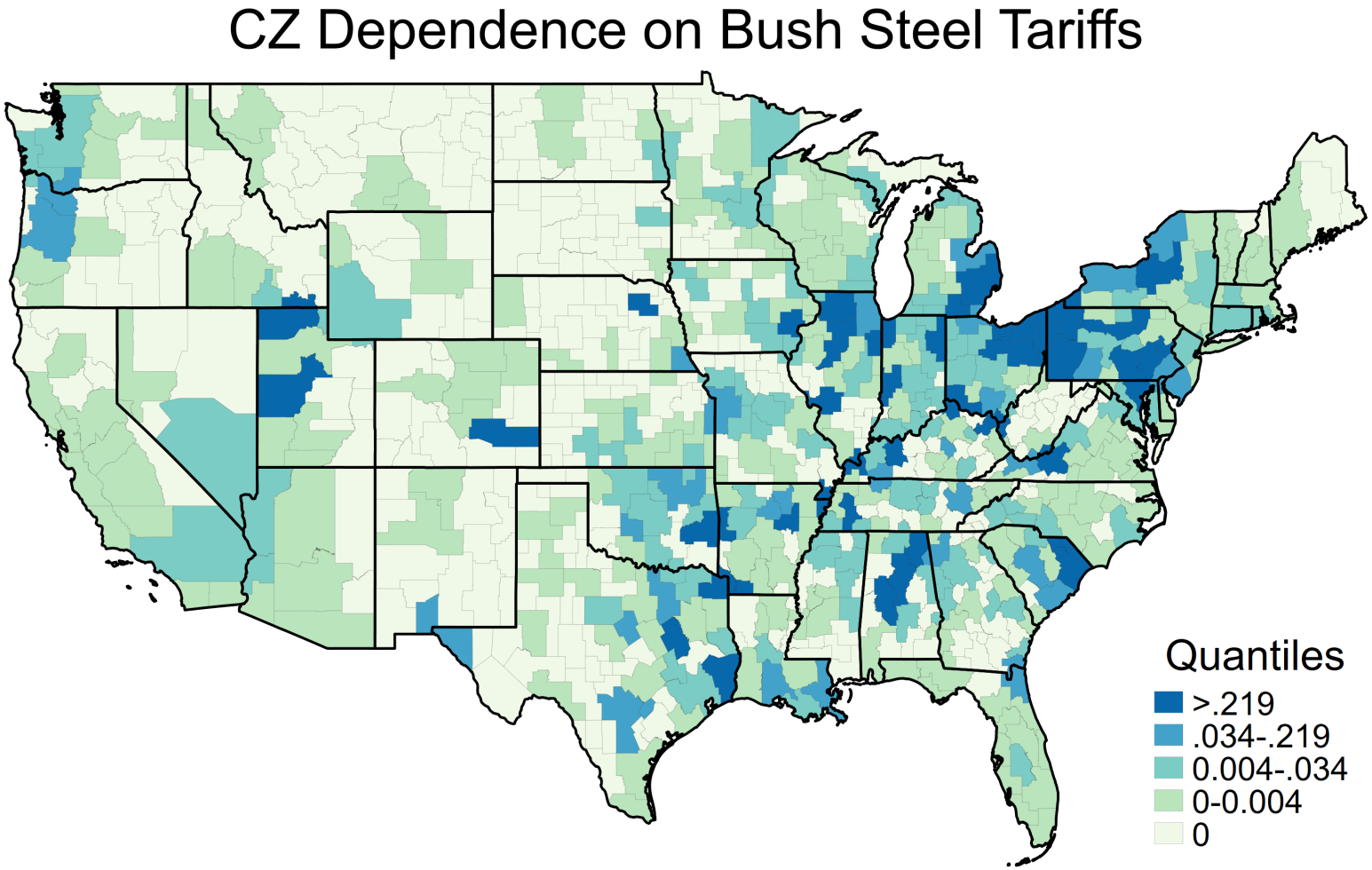
Main result II: elimination of industry tariffs

- Compare wage growth 1990-2000 for the following two workers
  - high school dropout in highest NAFTA exposed industry
    - Footwear, Oil & Gas, Capets, Plastics
  - high school dropout in lowest NAFTA exposed industry
    - Non-tradable industries (many types of services)
- Former has 17% pt lower wage growth
  - Eg: -5% for former, 12% for latter

Conclusion

- High school dropouts (often "blue collar" workers) hit hard in vulnerable locations & industries
- *Relative* to wage growth of HS dropout in *unaffected* industry and location
  - NAFTA reduces wage growth of HS dropout by...
  - 17% pts in most vulnerable *industry* but unaffected *location*
  - 8% pts in unaffected *industry* but most vulnerable *location*
  - 25% pts in most vulnerable *industry* and most vulnerable *location*
- Effects for other education levels
  - High school grads and some college: somewhat weaker
  - College grads: no effect

CZ dependence



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